

App Dev I — Project Report

Placement Portal Application

1. Student Details

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I am a student of the IIT Madras BS Degree programme with an interest in full-stack web development. I enjoy building practical applications that solve real-world organisational problems.

2. Project Details

Project Title: Placement Portal Application

Problem Statement

Institutes need an efficient system to manage campus recruitment involving companies, students, and placement drives. Manual processes — spreadsheets and emails — make it difficult to manage company approvals, track applications, prevent duplicate registrations, and maintain placement records. This project addresses these problems by building a role-based web application for Admin (placement cell), Company, and Student users.

Approach

The application was built using Flask with a Blueprint-based modular structure, separating concerns by role. Key design decisions included:

- Role-based access control using Flask-Login: each Blueprint (admin, company, student, auth) independently verifies the current user's role before processing requests.
- Database is created fully programmatically via SQLAlchemy models and seeded with an admin user through `create_db.py` — no manual DB creation.
- Business logic lives exclusively in the backend; Jinja2 handles templating with no JS dependencies for core features.
- A UniqueConstraint on (student_id, drive_id) in the Application table, combined with a route-level duplicate check, prevents multiple applications to the same drive.
- Cascade delete logic: removing a company cleans up all its drives and applications in a single transaction.

3. AI / LLM Declaration

I used Claude (Anthropic) as an AI assistant during this project. The extent of use is approximately 20–25%, primarily for:

- **HTML + Jinja2 (15%)**
Used for server-side template rendering, enabling dynamic content generation and seamless integration with backend logic.
- **Bootstrap / CSS (8%)**
Applied for styling and layout design, ensuring a responsive and visually consistent user interface. All final implementation, debugging, integration, and testing were done manually. The AI was used as a productivity aid and not as a substitute for original problem-solving.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask 3.0	Core backend web framework, Blueprint routing
Flask-Login 0.6	User session management and login_required guards
Flask-SQLAlchemy 3.1	ORM for database model definitions and queries
Werkzeug 3.0	Password hashing (generate/check_password_hash)
Jinja2 3.1	Server-side HTML templating
Bootstrap 5	Responsive frontend styling
SQLite	Lightweight local database (portal.db)
Python uuid	Auto-generation of student enrollment numbers

5. Database Schema / ER Diagram

Tables

- users — id, username, password (hashed), role (admin/student/company), is_active, is_blocked, created_at
- student_profiles — id, user_id (FK), full_name, email, phone, age, enrollment_no, roll_no, cgpa, placed, blocked, course_id (FK), department_id (FK), batch_id (FK), semester_id (FK), resume
- company_profiles — id, user_id (FK), registration_id, company_name, website, location, description, hr_name, hr_email, hr_phone, approved, rejected, blocked
- placement_drives — id, company_id (FK), job_title, job_description, eligibility, deadline, status (Pending/Approved/Rejected), closed, created_at
- applications — id, student_id (FK), drive_id (FK), applied_on, status (Applied/Shortlisted/Selected/Rejected) | UNIQUE(student_id, drive_id)
- courses — id, name
- departments — id, name
- batches — id, year
- semesters — id, number
- testimonials — id, user_id, role, rating, feedback, testimonial, is_approved, created_at

Relationships

- User 1 → 1 StudentProfile (user_id FK)
- User 1 → 1 CompanyProfile (user_id FK)
- CompanyProfile 1 → N PlacementDrive (company_id FK)
- StudentProfile 1 → N Application (student_id FK)
- PlacementDrive 1 → N Application (drive_id FK)
- Course / Department / Batch / Semester 1 → N StudentProfile (FK references)

- Placement drive lifecycle: Pending → Approved/Rejected → Closed
- Student application with duplicate prevention (DB UniqueConstraint + route check)
- Company shortlisting workflow: Applied → Shortlisted → Selected/Rejected; selecting a student auto-marks them as placed
- Admin dashboard with 6 real-time statistics
- Advanced student search and filter (name, email, roll/enrollment, course, dept, batch, semester, min CGPA, placed status)
- Resume upload: PDF-only, 2 MB limit, secure filename handling
- Cascade delete for companies (removes all drives and applications)

Additional Features (Beyond Core Requirements)

- Testimonials system: students and companies submit feedback; admin moderates and publishes on the landing page
- Master data management: admin can add/delete Courses, Departments, Batches, and Semesters
- Auto-generated UUID-based enrollment number assigned at student registration
- Public landing page with portal overview and approved testimonials

8. Video Presentation

Drive Link: https://drive.google.com/file/d/1WMRsdHIR1Lufc-Bxkj_GXZPxOpmi3McE/view?usp=drive_link